Choosing Table Primary Key

Reminder: What is a primary key?

- Superkey:
  - A set of attributes such that …
  - If tuples have the same superkey values,
  - Then tuples are the same.
- Key: a minimal superkey
- Primary key: a chosen key

Pragmatic goals for PK

- Unique
- Exists/known
- Immutable
- Simple/compact

Two design questions

1. Natural vs. synthetic?
2. How to choose among multiple natural keys?

Natural vs. synthetic?

Natural

- Meaningful data already included in DB needs

Synthetic

- Meaningless
- Made up for internal DB use

Table	Primary key???
Person	SSN, thumbprint
Course	Dept code + number
Team player	Jersey number
Campus building	Name, abbreviation
Library book	ISBN

Main tradeoffs

	Natural	Synthetic
Already need data	Yes	No
Unique	Usually	Yes
Exists/known	Usually	Yes
Immutable	Usually	Almost always
Simple/compact	Usually	Yes

Natural vs. Synthetic: My recommendation

Use synthetic keys only as substitutes for complex natural keys.



How to choose among natural keys?

Pick which best meets goals:

- Unique
- Exists/known
- Immutable
- Simple/compact

How to generate synthetic keys?

- `SMALLSERIAL`, `SERIAL`, `BIGSERIAL` types
  - Auto-incrementing integers, fast
  - Insufficient for concurrent systems

- `UUID` type & `uuid_generate_v4()`
  - Unique even in concurrent systems